

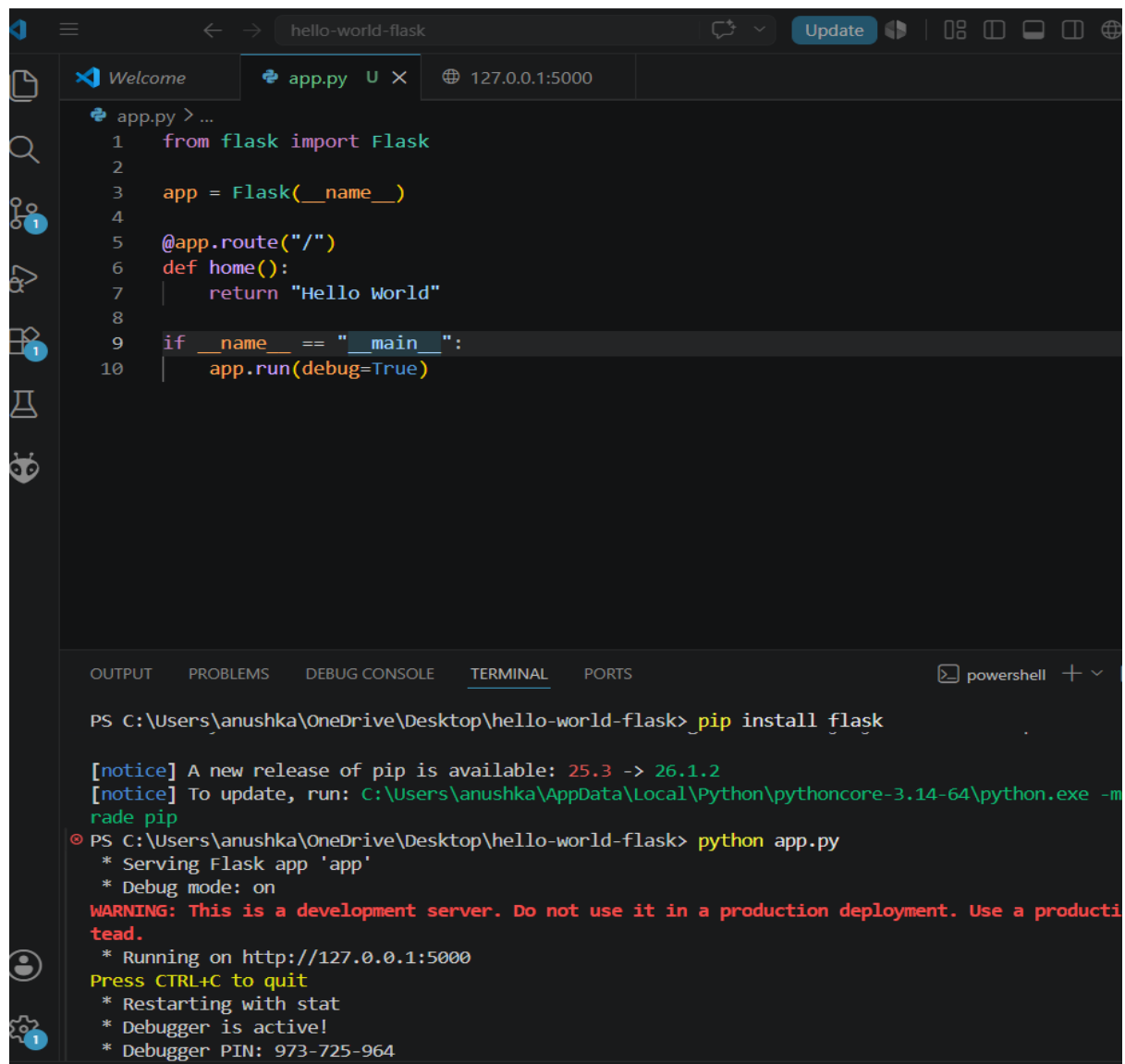
Name: Anushka Desai

PRN:23070122035

1. Git Workflow & Collaboration (4 Marks)

- **Task 1.1:** Initialize a new Git repository for a simple "Hello World" Python Flask application (provided to you). Add the initial code and commit it to the main branch. (1 Mark)
- **Task 1.2:** Create a new branch named feature/user-auth. Make a small modification (e.g., add a new print statement) in this branch, commit the changes, and push the branch to a remote repository (e.g., GitHub/GitLab). (1.5 Marks)
- **Task 1.3:** Simulate a conflict: On the main branch, modify the *same line* you changed in feature/user-auth. Commit this change. Now, attempt to merge feature/user-auth into main. Resolve the merge conflict manually and commit the resolution. Push the updated main branch. (1.5 Marks)

1.app.py

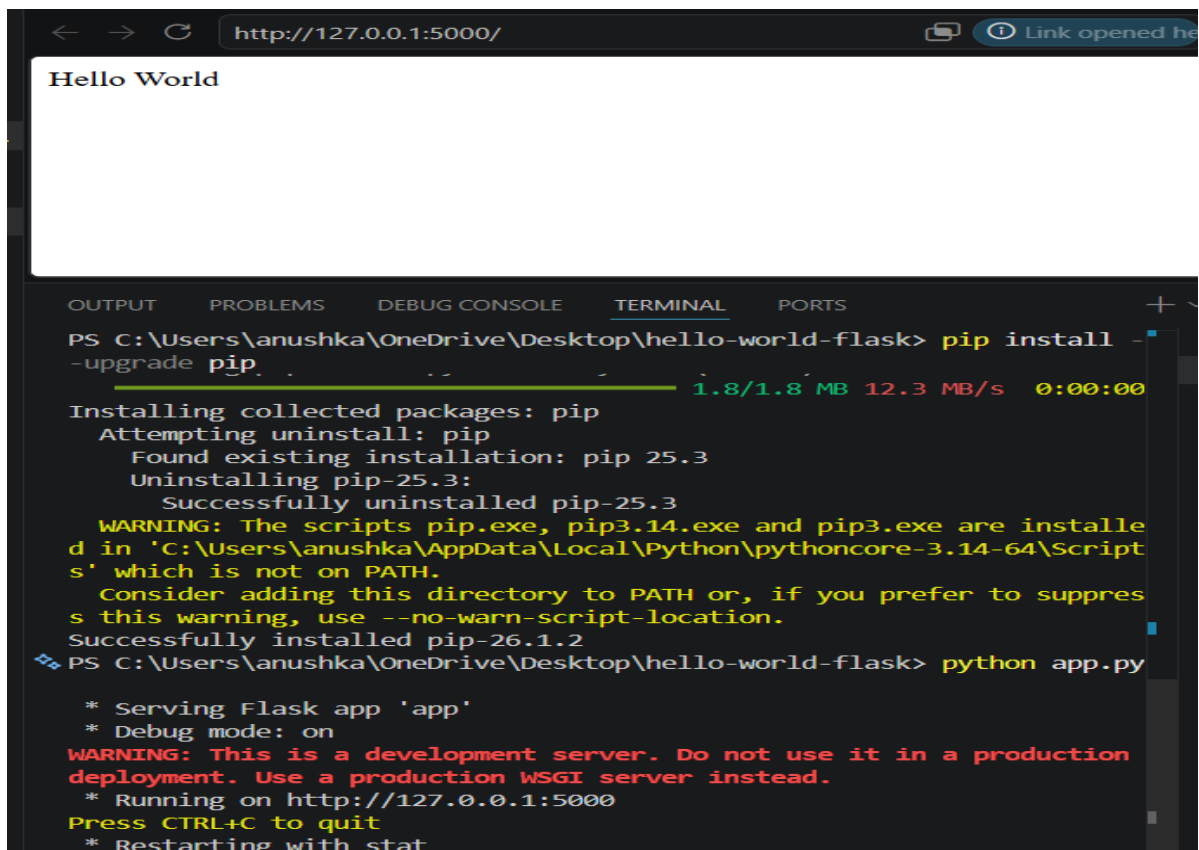


The screenshot shows the Visual Studio Code editor with a file named `app.py` open. The code is a simple Flask application that returns "Hello World" at the root URL. Below the editor, the integrated terminal shows the command `pip install flask` being executed, followed by `python app.py`. The terminal output indicates that the Flask app is running on `http://127.0.0.1:5000` in debug mode.

```
app.py > ...
1  from flask import Flask
2
3  app = Flask(__name__)
4
5  @app.route("/")
6  def home():
7      return "Hello World"
8
9  if __name__ == "__main__":
10     app.run(debug=True)
```

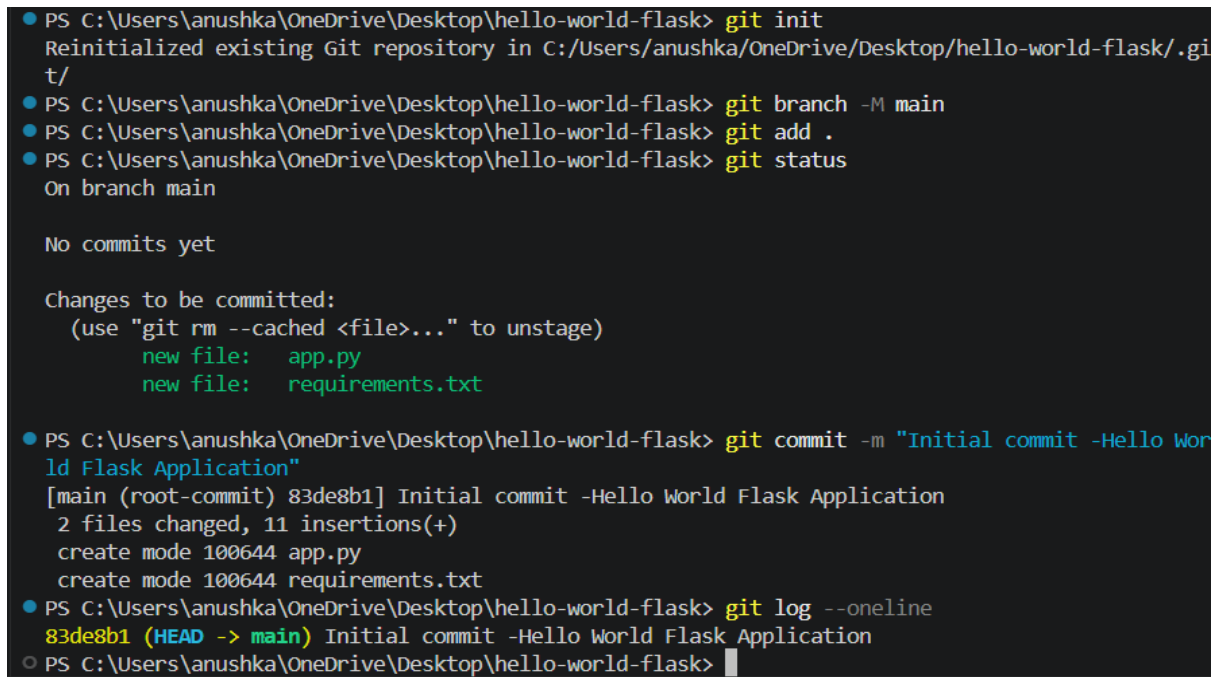
```
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> pip install flask
[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: C:\Users\anushka\AppData\Local\Python\pythoncore-3.14-64\python.exe -m
rade pip
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a producti
thead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 973-725-964
```

2. "Hello World" Output in Browser and Flask Server Running



```
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> pip install --upgrade pip
1.8/1.8 MB 12.3 MB/s 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.3
    Uninstalling pip-25.3:
      Successfully uninstalled pip-25.3
  WARNING: The scripts pip.exe, pip3.14.exe and pip3.exe are installed in 'C:\Users\anushka\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  Successfully installed pip-26.1.2
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
```

3. Git Repository Initialization, Initial Commit, and Push to Main Branch



```
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git init
Reinitialized existing Git repository in C:/Users/anushka/OneDrive/Desktop/hello-world-flask/.git/

PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git branch -M main
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git add .
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   app.py
    new file:   requirements.txt

PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git commit -m "Initial commit -Hello World Flask Application"
[main (root-commit) 83de8b1] Initial commit -Hello World Flask Application
 2 files changed, 11 insertions(+)
 create mode 100644 app.py
 create mode 100644 requirements.txt

PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git log --oneline
83de8b1 (HEAD -> main) Initial commit -Hello World Flask Application

PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask>
```

4. Creating feature/user-auth Branch, Committing Changes, and Pushing to Remote Repository

```

create mode 100644 requirements.txt
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git log --oneline
83de8b1 (HEAD -> main) Initial commit -Hello World Flask Application
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git remote add origin https://github.com/AnushkaDesai13/hello-world-flask.git
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 409 bytes | 204.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AnushkaDesai13/hello-world-flask.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask>

```

5. Updating Main Branch and Generating Merge Conflict During Branch Merge

```

hello-world-flask
Update

OUTPUT  PROBLEMS  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git commit -m "Initial commit -Hello World Flask Application"
create mode 100644 requirements.txt
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git log --oneline
83de8b1 (HEAD -> main) Initial commit -Hello World Flask Application
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git remote add origin https://github.com/AnushkaDesai13/hello-world-flask.git
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 409 bytes | 204.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AnushkaDesai13/hello-world-flask.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git checkout -b feature/user-auth
Switched to a new branch 'feature/user-auth'
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git add .
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git commit -m "Added user authentication feature"
[feature/user-auth 0493654] Added user authentication feature
1 file changed, 1 insertion(+)
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push -u origin feature/user-auth
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 366 bytes | 366.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/user-auth' on GitHub by visiting:
remote:   https://github.com/AnushkaDesai13/hello-world-flask/pull/new/feature/user-auth
remote:
To https://github.com/AnushkaDesai13/hello-world-flask.git
 * [new branch]      feature/user-auth -> feature/user-auth

```

```

nothing to commit, working tree clean
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push
Everything up-to-date
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git log --oneline
0493654 (HEAD -> feature/user-auth, origin/feature/user-auth) Added user authentication feature
83de8b1 (origin/main, main) Initial commit -Hello World Flask Application
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> type app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    print("User Authentication Feature Added")
    return "Hello World"

if __name__ == "__main__":
    app.run(debug=True)
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git add .
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git commit -m "Updated main branch"
[main 2861546] Updated main branch
 1 file changed, 1 insertion(+), 1 deletion(-)
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 332 bytes | 332.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/AnushkaDesai13/hello-world-flask.git
 83de8b1..2861546  main -> main
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git merge feature/user-auth
Auto-merging app.py
CONFLICT (content): Merge conflict in app.py
Automatic merge failed; fix conflicts and then commit the result.
• PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git status
On branch main
Your branch is up to date with 'origin/main'.

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)

```

6. Updating Main Branch and Generating Merge Conflict During Branch Merge

```
hello-world-flask
Welcome app.py 8,! × requirements.txt 127.0.0.1:5000
app.py > ...
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def home():
7     <<<<<<< HEAD (Current Change)
8     return "Hello World from Main Branch"
9     =====
10    print("User Authentication Feature Added")
11    return "Hello World"
12    >>>>>>> feature/user-auth (Incoming Change)
13
14 if __name__ == "__main__":
15     app.run(debug=True)
```

```
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git merge feature/user-auth
● PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git status
On branch main
Your branch is up to date with 'origin/main'.

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
● PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git add app.py
● PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git commit -m "Resolved merge conflict"
[main 20b2c7d] Resolved merge conflict
● PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 390 bytes | 390.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/AnushkaDesai13/hello-world-flask.git
 2861546..20b2c7d  main -> main
○ PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask>
```

7. Final Git log showing complete repository history.

```
PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git status

nothing added to commit but untracked files present (use "git add" to track)
● PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git clean -fd
● PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask> git log --oneline --graph --all
* 20b2c7d (HEAD -> main, origin/main) Resolved merge conflict
| \
| * 0493654 (origin/feature/user-auth, feature/user-auth) Added user authentication feature
* | 2861546 Updated main branch
| /
* 83de8b1 Initial commit -Hello World Flask Application
○ PS C:\Users\anushka\OneDrive\Desktop\hello-world-flask>
```